

ACUTE KIDNEY INJURY AND CHRONIC KIDNEY DISEASE

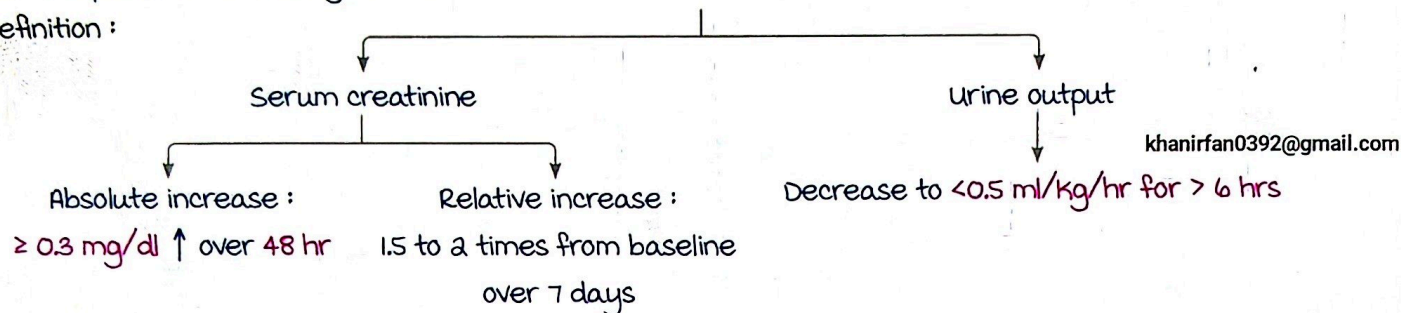
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Acute Kidney Injury (AKI)

00:00:15

Acute impairment of kidney function.

Definition :



AKI staging :

Stage	Serum creatinine	urine output
1	↑ of ≥ 0.3 mg/dl over 48 hrs OR ≥ 1.5 to 2 fold from baseline in last 7 days	< 0.5 ml/kg/hr for >6hrs
2	1 to 2 to 3 fold from baseline.	< 0.5 ml/kg/hr for ≥ 12 hrs
3	• 3x ↑ from baseline OR, • Absolute serum creatinine ≥ 4mg/dl OR, • Patient undergoing dialysis OR, • eGFR <35 ml/min/1.73 m ² .	< 0.3 ml/kg/hr for ≥ 24 hrs OR Anuria ≥ 12 hrs

Schwartz formula :

$$eGFR = \frac{\text{Length of child (cm)} \times K \text{ (Constant)}}{\text{Serum creatinine (mg/dl)}}$$

Where K= 0.413 in all age groups.

Drawbacks of using S. creatinine for staging :

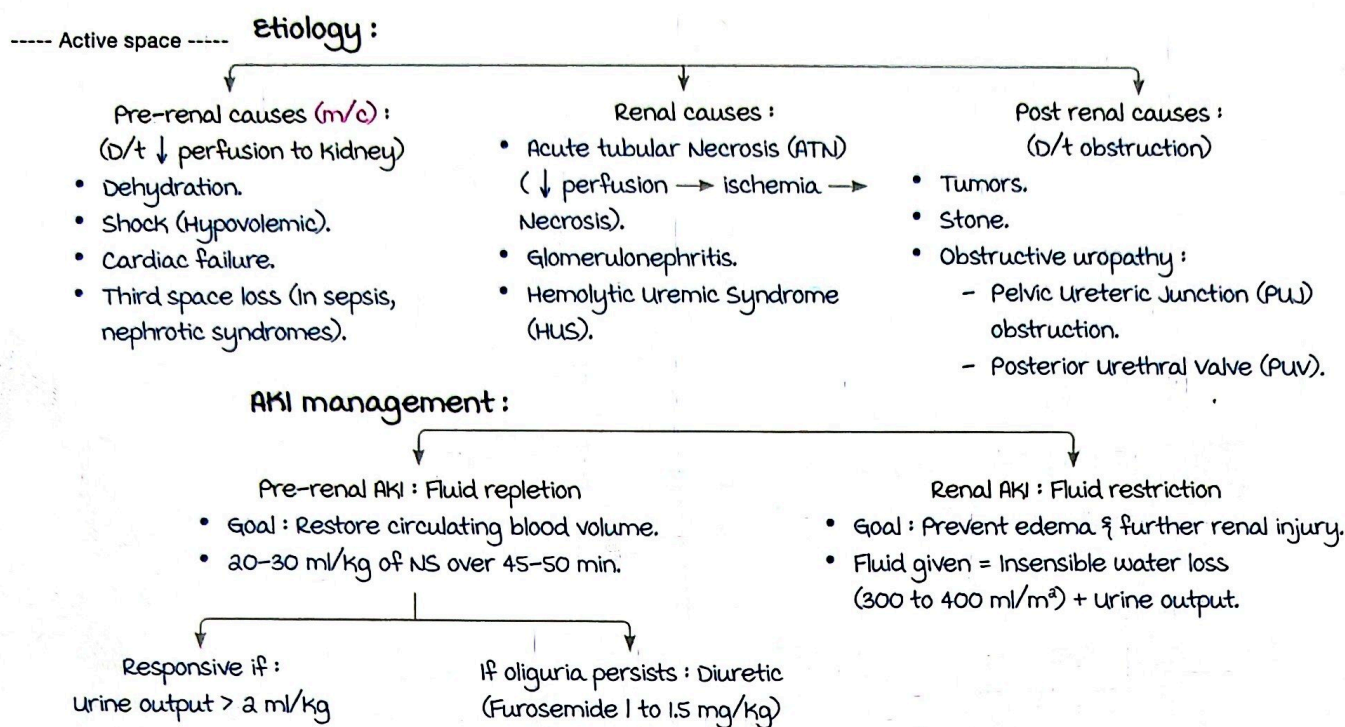
1. Age-dependent : value of S.creatinine varies according to muscle mass.
2. Delayed rise in S. creatinine following AKI.
3. Unreliable d/t absence of previous/base value.

Newer biomarkers : To overcome drawbacks of s.creatinine.

1. Cystatin C.
2. Neutrophil Gelatinase Associated Lipocalin (NGAL).
3. Kidney Injury molecule- 1 (Kim-1).

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Complications	management	
Anemia	D/t hemodilution : usually resolves on management with fluids	D/t blood loss : Packed Red Blood cell (PRBC) transfusion
metabolic acidosis	IV sodium bicarbonate	
Hyponatremia	Asymptomatic	Symptomatic
	Resolves on fluid management	3 % saline (Hypertonic)
Hyperkalemia (S.K ⁺ > 5.5 mEq/L)	↓ K ⁺ levels	
	Shift K ⁺ intracellularly	↑ K ⁺ excretion (Binders)
	• Insulin-glucose drip. • NaHCO₃. • Neb. Salbutamol.	• K⁺ binding resins. • Kayexalate (Sodium polystyrene sulfate).
	If refractory → Dialysis	
	High K ⁺ (> 7 mEq/L) → Conduction abnormalities. Slow IV infusion of 10% Calcium gluconate with cardiac monitoring (Fast infusion C/I d/t effect on conduction)	

HEMOLYTIC UREMIC SYNDROME :**Types :**

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D+ (m/c)

(Associated with diarrhoea)

D/t shiga-toxin produced by :

- Shigella dysenteriae type I.
- Enterohemorrhagic E. Coli (EHEC) :

Strain O157 : H7

D -

(Not associated with diarrhoea)

Defect in alternate complement pathway

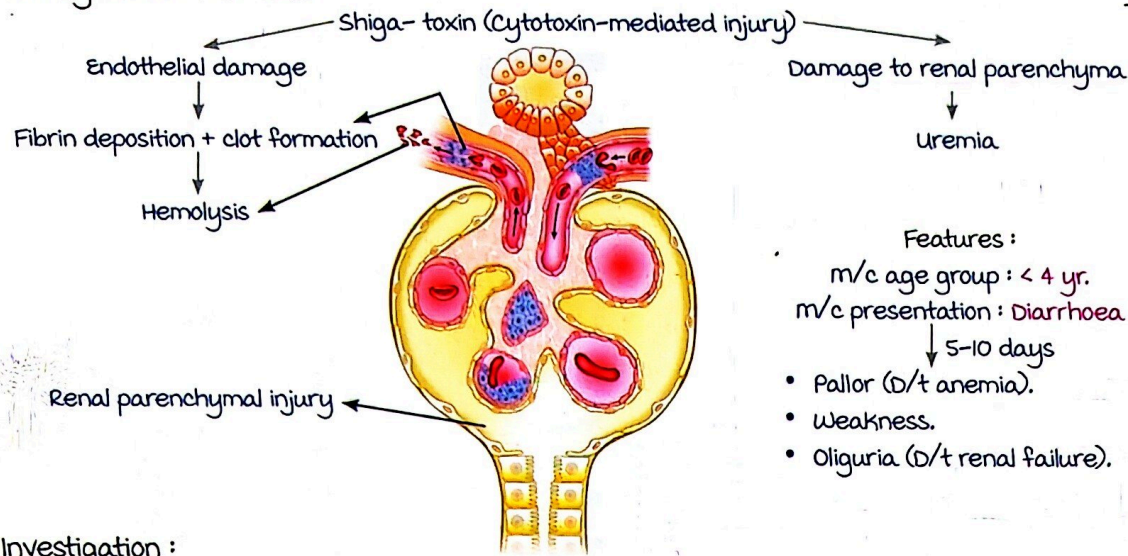
↓ Factor H

↓

↑↑ activation of complement pathway

(↑ activation of C5)

Pathogenesis of D+ HUS :

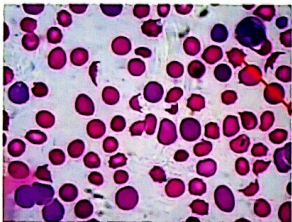


Features :

- m/c age group : < 4 yr.
m/c presentation : Diarrhoea
↓ 5-10 days
- Pallor (D/t anemia).
 - Weakness.
 - Oliguria (D/t renal failure).

Investigation :

a. Peripheral smear



b. Blood investigations :

- ↑ Blood urea.
- ↑ S. creatinine.

Schistocyte (Lysed RBCs)

c. COOMBS test :

Negative
(No anti bodies)

d. PCR/ELISA on stool sample

for Shiga-toxin. definitive
test.

management :

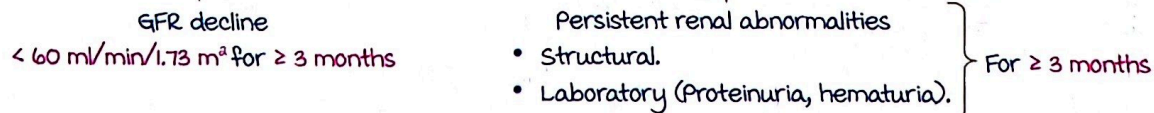
D+ HUS	D- HUS
• Usually self-resolving → Supportive. • C/I : Antibiotics (Lysis of bacteria → ↑ toxins to blood).	• CS inhibitors : Blocks complement activation. • Eculizumab and Ravulizumab.
If severe → Dialysis	

Chronic Kidney Injury (CKD)

00:28:12

Chronic decline in renal function.

Definition :



STAGING OF CKD :

Stage	GFR (In ml/min/1.73 m ²)
1 (Hyperfiltration)	>90 (D/t compensation by hypertrophy & surviving nephrons)
2	60-89
3	30-59
4	15-30
5	< 15 or if patient is on dialysis

Feedback

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ETIOLOGY :**Obstructive uropathy :**

- Stones.
- Tumors.
- Strictures.

Reflex nephropathy :

- Vesico-ureteric reflex causes scars in kidney.

Chronic glomerulonephritis :

- Ig A nephropathy (Berger's disease)!

Congenital anomalies (m/c)**MANAGEMENT :**

Cause	management
Structural anomaly (Reversible)	Surgical correction
Irreversible	medical management : • No diet restriction (including proteins) : To promote growth. • Intake of polyunsaturated fatty acids (PUFAs) and complex carbohydrates.

management of complications :

Complication	Pathogenesis	management
Anemia	$GFR < 30 \text{ ml/min/1.73 m}^2$ ↓ Erythropoietin	Darbopoetin : SC injection (Recombinant erythropoietin)
Mineral bone disease	1. ↓ levels of 1- α hydroxylase. ↓ 2. ↓ active vit D levels (1, 25-(OH) $_2$ D) ↓ 3. Hypocalcemia ↓ 4. Secondary hyperparathyroidism (↑ PTH) ↓ 5. Bone resorption	Calcitriol supplementation Calcium supplementation Resolves upon treatment of 2 and 3
	Advanced CKD ↓ PO_4^{3-} excretion ↓ Hyperphosphatemia	Phosphate binders : • Ca based : Acetate/citrate. • Non-Ca based : Sevelamer.
Short stature	-	Recombinant growth hormone (rGH)
Hypertension (HTN)	-	Lifestyle management : • Salt restriction (< 2 g/day). • Exercise and weight management.
		medical management : • Angiotensin Converting Enzyme (ACE) inhibitors. • Angiotensin receptor blockers (ARB).

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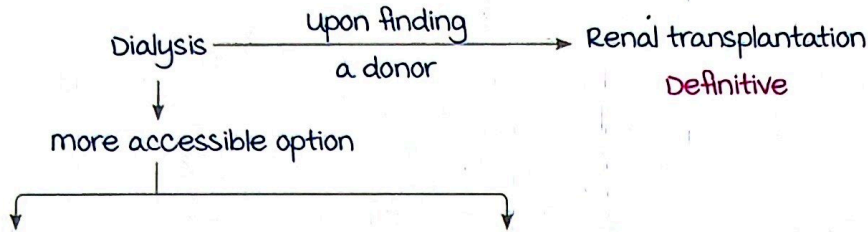
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Kidney Replacement Therapy (KRT):

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For progressive CKD refractory to medication.

Initiation at $\text{GFR} < 12 \text{ ml/min/1.73 m}^2$.

Peritoneal dialysis:

- Home set-up.
- Approx 12 hr/day.
- Better HTN control.

Hemodialysis:

- Hospital set-up.
- Approx 3 times/week for 3-4 hrs each.